

# Asymmetry in Stock Comovements: An Entropy Approach

**Lei Jiang**

Tsinghua University

**Ke Wu**

Renmin University of China

**Guofu Zhou\***

Washington University in St. Louis

First Version: June 2014

Current Version: June 2017

---

\*Corresponding author. Send correspondence to Guofu Zhou, Olin School of Business, Washington University in St. Louis, St. Louis, MO 63130; e-mail: [zhou@wustl.edu](mailto:zhou@wustl.edu); phone: 314-935-6384. We are grateful to Vikas Agarwal, Doron E. Avramov, Jonathan Brogaard, Jeffrey A. Busse, Hui Chen, Tarun Chordia, Lauren H. Cohen, Philip H. Dybvig, Andrey Ermolov, Ping He, Zhiguo He, David Jacho-Chavez, Ying Jiang, Raymond Kan, Omesh Kini, Junghoon Lee, Tingjun Liu, Esfandiar Maasoumi, Michael R. Powers, Jun Tu, Aurelio Vasquez, Baozhong Yang, and seminar participants at Cornerstone Research, Case Western Reserve University, Emory University, Georgia State University, Harbin Institute of Technology, ITAM, Peking University, Washington University in St. Louis, 2014 and 2015 Tsinghua Finance Workshops, 2015 China Finance Review International Conference, 2015 Five-Star Workshop in Finance, 2015 China International Conference in Finance, and 2017 Midwest Finance Association Annual Meeting for numerous helpful comments and especially to Jennifer Conrad (the editor) and an anonymous referee for their many insightful and detailed comments that have substantially improved the paper. Lei Jiang gratefully acknowledges financial support from the AXA Research Fund, the Initiative Research Program supported by Tsinghua University (20151080398), the National Science Foundation of China (71572091), and the Tsinghua National Laboratory for Information Science and Technology. The authors thank Jing Ding and Jinyu Liu for excellent research assistance.

# Asymmetry in Stock Comovements: An Entropy Approach

## Abstract

We provide an entropy approach for measuring asymmetric comovement between the return on a single asset and the market return. This approach yields a model-free test for stock return asymmetry, generalizing the correlation-based test proposed by Hong, Tu, and Zhou (2007). Based on this test, we find that asymmetry is much more pervasive than previously thought. Moreover, our approach also provides an entropy-based measure of downside asymmetric comovement. In the cross-section of stock returns, we find an asymmetry premium: high downside asymmetric comovement with the market indicates higher expected returns.

*JEL* classifications: C12, C15, G12

Keywords: Asymmetric comovement, entropy, asset pricing

## I. Introduction

A number of studies have found asymmetries in asset returns with respect to the market. Ball and Kothari (1989), Schwert (1989), Conrad, Gultekin, and Kaul (1991), Kroner and Ng (1998), Cho and Engle (1999), Bekaert and Wu (2000), Ang and Chen (2002), Bae, Karolyi, and Stulz (2003), and Ang, Chen, and Xing (2006), among others, document asymmetries in the covariances, correlations, and betas of stock returns. Recently, Lettau, Maggiori, and Weber (2014) have also found asymmetric betas in currency portfolios. Such asymmetries in asset returns are important for both portfolio selection and risk management, since diversification and hedging strategies rely critically on the ways in which asset returns are dependent upon each other. While earlier tests of symmetry in stock market are model-dependent, Hong, Tu, and Zhou (2007) propose a model-free test (henceforth the HTZ test). This test, despite its great potential, has some weaknesses. First, it does not address asymmetry beyond the second moment and hence cannot capture the full dependence structure for non-normal distributions; this is a concern because realized stock returns are well known to be non-normally distributed (e.g., Zhou, 1993; Ang and Chen, 2002). Second, the test has low power and fails to detect asymmetry in many portfolios. For example, no asymmetry is found in book-to-market portfolios, which appears inconsistent with the results of Ang and Chen (2002).

In this paper, we propose an entropy measure to quantify asymmetry in the joint distribution of an individual stock return and the market return, and then extend it to measure their downside asymmetric comovement. Based on the entropy asymmetry measure, we provide a new model-free test to determine whether or not asymmetric comovement exists in the stock market. Statistically, the test statistic is a bivariate normalized version of the entropy measure originally proposed by Granger, Maasoumi, and Racine (2004), which has been widely applied in econometrics (see, e.g., Maasoumi and Racine, 2002; Racine and Maasoumi, 2007). The entropy measure, based

on the joint probability density functions, summarizes all the information in the joint distribution, thus capturing the general asymmetric dependence structure that exists in all moments beyond the second.

With Monte Carlo simulations, we find that the entropy test has correct empirical size and is generally more powerful than the HTZ test. Using portfolios sorted by size, book-to-market ratio, or prior (2- to 12-month) returns, we find that the entropy test detects statistically significant asymmetry in all of them. For example, in contrast to the HTZ test, which fails to reject symmetry for all deciles of book-to-market portfolios, the entropy test rejects symmetry in seven out of ten portfolios at the 10% significance level using a single exceedance level.<sup>1</sup>

What are the asset pricing implications of asymmetric comovements on the cross-section of stock returns? To address this question, we extend the entropy asymmetry measure to a measure of downside asymmetric comovement. Ang, Bekaert, and Liu (2005) show that downside asymmetry risk matters to investors with disappointment aversion (DA) preferences, and Dahlquist, Farago, and Tédongap (2017) further develop portfolio strategies under asymmetry. Since investors dislike stocks with downside asymmetric comovement with the market, we expect a risk premium for such stocks. Whether or not the size of this premium is of economic significance is an empirical question that we investigate here.

By sorting stocks into quintile portfolios by their prior downside asymmetry, we find that the return difference between the highest quintile and lowest quintile can be as high as 4.5% per annum, which is both economically and statistically significant.<sup>2</sup> Since this difference in returns is a compensation for downside asymmetry risk, it is not surprising that it cannot be explained by standard factor models such as the four-factor model in Carhart (1997). Moreover, the risk

---

<sup>1</sup>At the 5% level, it rejects two out of the ten using the single exceedance level; with multiple exceedance levels, symmetry is rejected in six out of the ten portfolios.

<sup>2</sup>To avoid the look-ahead bias, we sort portfolios based on predetermined variables only. If the contemporaneous downside asymmetry measure were used, the return spread would more than double.

premium is time-varying, rising during periods of low investor sentiment, low market volatility and high market liquidity.

We also compare our entropy-based downside asymmetry measure with downside beta (Ang, Chen, and Xing, 2006) and coskewness (Harvey and Siddique, 2000). The downside beta captures a stock's systematic risk exposure to the downside market, while coskewness describes the comovement of an asset return with the square of the market return. Differences in returns between stocks sorted by either of these two measures are much smaller and can be fully explained by Carhart (1997)'s four-factor model. Interestingly, the return spread between stocks sorted by a downside asymmetry risk measure based on the HTZ test statistic earns a statistically significant risk premium of 2.93% per annum. However, the entropy-based measure yields an approximately 50% higher risk premium. Since existing measures are based on the second and third moments only, the results show that our asymmetry measure, based on the entire joint density function, is indeed relevant in practice.

Backus, Boyarchenko, and Chernov (2016) and Chabi-Yo and Colacito (2016), motivated from different applications, propose two alternative co-entropy measures that capture nonlinear co-dependence between two random variables beyond the Pearson correlation. Interestingly, when we apply their measures to our sample, we also find statistically significant risk premiums greater than those from both the downside beta and the downside HTZ test statistic. These results suggest that downside asymmetry in nonlinear co-dependence beyond the simple correlation is important and commands a substantial and statistically significant premium in the cross-section of stocks, robust to various entropy-based measures.

There are several recent studies emphasizing on tail risk and cross-sectional asset pricing implications. For example, Kelly and Jiang (2014) propose a left tail risk factor and find that the higher exposure to the tail risk, the higher the expected returns. Chabi-Yo, Ruenzi, and Weigert (2014) examine copula-based left tail dependence and find that it has explanatory power on the

cross-section of stock returns. In contrast, we focus on the difference between the left and right tails rather than their absolute magnitude. In other words, no matter how fat the left tail of the joint distribution of a stock return and the market return may be, we do not consider the distribution asymmetrical as long as it has an equally fat right tail. Our paper also differs from studies of the asset pricing effect of asymmetry in individual stock distributions or asymmetry in the distributions of macroeconomic variables. For example, Conrad, Dittmar, and Ghysels (2013) and Amaya et al. (2015) find evidence that stocks with right-skewed return distributions have lower cross-sectional average returns. Colacito et al. (2016) introduce skewness into consumption-based asset pricing models and provide empirical evidence that the skewness of analysts' forecasts for GDP growth predicts equity returns. Unlike these studies, we focus on the asymmetry in comovements of individual stock returns with the market return.

The remainder of the paper is organized as follows. Section II introduces the entropy test for joint distribution asymmetry. Section III examines the test's size and power using Monte Carlo simulations. Section IV applies the entropy test to investigate asymmetry in common portfolios. Section V analyzes the impact of asymmetry on the cross-section of stock returns. Section VI concludes the paper.

## **II. Tests of Asymmetry**

In this section, we first briefly review standard symmetry tests based on exceedance correlations, then extend the concept to more general asymmetry, and finally provide our entropy test.

### **A. Asymmetric Correlation**

Ang and Chen (2002) and Hong, Tu, and Zhou (2007) provide tests of asymmetry for a stock's comovement with the market, but both of these tests address only exceedance correlation asymmetry.

To see why, let  $x$  and  $y$  be pairs of stock return series, and  $\tilde{x}$  and  $\tilde{y}$  be their standardizations. Both tests rely on exceedance correlations, that is, the conditional correlations calculated when both individual stock and market returns are below or above certain exceedance levels. The exceedance correlations at some given level  $c$  are defined as

$$(1) \quad \rho^+(c) = \text{corr}(\tilde{x}, \tilde{y} | \tilde{x} > c, \tilde{y} > c),$$

$$(2) \quad \rho^-(c) = \text{corr}(\tilde{x}, \tilde{y} | \tilde{x} < -c, \tilde{y} < -c), \quad \forall c \geq 0.$$

Clearly, both  $\rho^+(c)$  and  $\rho^-(c)$  measure conditional correlations between two return series conditioning on whether both series are above or below a certain exceedance level  $c$ . The null hypothesis of interest is

$$H_0 : \quad \rho^+(c) = \rho^-(c), \quad \text{for all } c \geq 0.$$

If the null is true, the exceedance correlations are symmetric.

Ang and Chen's (2002) test addresses whether the asymmetric correlations observed in the data can be explained by a given model. Therefore, the test is model-dependent, and its results rely on the choice of model. One weakness associated with such a test is that rejection of the symmetry hypothesis could be a result of rejection of the given model.

To overcome this weakness, Hong, Tu, and Zhou (2007) propose a model-free test. Their test statistic is defined as

$$(3) \quad J_\rho = T(\hat{\rho}^+ - \hat{\rho}^-)' \hat{\Omega}^{-1} (\hat{\rho}^+ - \hat{\rho}^-),$$

where  $T$  is the sample size,  $\hat{\rho}^+$  and  $\hat{\rho}^-$  are  $m \times 1$  vectors of sample exceedance correlations, and  $\hat{\Omega}$  is a consistent estimator of the covariance matrix of  $\sqrt{T}(\hat{\rho}^+ - \hat{\rho}^-)$ . Under the null of symmetric correlations and certain regularity conditions, the test has a simple asymptotic chi-

square distribution,  $J_\rho \xrightarrow{d} \chi_m^2$ . The test answers the question of whether or not asymmetric correlations exist in the data at all. In other words, if the test rejects the null, no distributions with symmetric correlations can fit the data well.

## B. Asymmetric Comovement

The methodologies presented in the previous subsection use correlations to test for asymmetric comovement in bivariate stock returns. However, it is well known that a correlation coefficient measures only linear dependence and cannot reflect a dependence structure beyond the second moment. In general, a zero correlation does not imply independence if the joint distribution is non-normally distributed. Hence, tests based on correlations ignore possible higher-order co-dependence. Because research has established that financial and macroeconomic time series are usually leptokurtic and have heavy tails (e.g., Zhou, 1993; Cont, 2001; Ang and Chen, 2002; Bekaert and Engstrom, 2016),<sup>3</sup> studying asymmetric comovement in asset returns requires a test that involves all higher-order moments.

Since the joint density function uniquely defines a joint distribution, directly testing for asymmetry in the joint probability density function certainly involves all higher-order moments. We start with a univariate test of distributional asymmetry proposed by Racine and Maasoumi (2007) and extend it to the case of joint distributions.

Analogous to the exceedance correlations, we define exceedance densities as

$$(4) \quad f^+(c, \tilde{x}, \tilde{y}) = f(\tilde{x}, \tilde{y} | \tilde{x} > c, \tilde{y} > c),$$

$$(5) \quad f^-(c, \tilde{x}, \tilde{y}) = f(-\tilde{x}, -\tilde{y} | \tilde{x} < -c, \tilde{y} < -c),$$

where  $f^+(c, \tilde{x}, \tilde{y})$  is the joint probability density function of standardized returns in the ranges of

---

<sup>3</sup>Beyond returns, Bekaert, Engstrom, and Ermolov (2015) provide an asymmetric model of volatility.

$\tilde{x} > c$  and  $\tilde{y} > c$ .<sup>4</sup> Restrictions to these ranges, where both returns are above or below a certain exceedance level, follow from the work of Ang and Chen (2002) and Hong, Tu, and Zhou (2007) to capture the comovements in lower and upper tails of the joint return distribution. Since  $\tilde{x}_t$  and  $\tilde{y}_t$  are standardized to have zero means,  $f^-(c, \tilde{x}, \tilde{y})$  denotes the joint probability density function of the rotated return series around the mean in the ranges of  $\tilde{x} < -c$  and  $\tilde{y} < -c$ . If the joint distribution is truly symmetric, then the two densities should be the same almost everywhere. Intuitively, the distance between the two density functions reflects the degree of asymmetry of the joint return distribution. Hence, our null hypothesis for testing symmetric comovement is

$$(6) \quad H_0 : \quad f^+(c, \tilde{x}, \tilde{y}) = f^-(c, \tilde{x}, \tilde{y}), \quad \text{for all } (c, \tilde{x}, \tilde{y}) \in \mathbb{R}_+ \times \mathbb{R}^2.$$

If this null hypothesis is rejected, then the data must possess asymmetry, since the density function after a certain rotation differs from the original density function.

[Insert Figure 1 about here]

To explain the intuition behind this proposed entropy test, we give an illustrative example. Consider two random variables  $X$  and  $Y$ , which may be regarded as the standardized returns on a stock and on the market. Let their joint distribution be characterized by the contour plots of Figure 1. In the first quadrant,  $X$  and  $Y$  are drawn from a joint normal distribution with means of 3.5, with unit variances and a Pearson correlation of 0.5, while in the third quadrant, they are drawn independently from two joint normal distributions with equal probability of 0.5. Both joint normal distributions have means of  $-3.5$  and unit variances, but one of them has zero correlation and the other one has correlation coefficient equal to 1. Then by construction, we know that the Pearson correlation coefficient is exactly the same, 0.5, in both quadrants. However, it is clear from Figure 1 that the co-dependence between  $X$  and  $Y$  looks quite different across the quadrants. This is because

---

<sup>4</sup>We may consider  $\tilde{x}$  as return to a single stock and  $\tilde{y}$  as the market return.

Pearson's correlation can only pick up the linear co-dependence between two random variables, and thus becomes an inaccurate measure of co-dependence in the presence of non-linearities.

Consider the point  $(X_0, Y_0)$  in the first quadrant and its associated reflection point around the means  $(-X_0, -Y_0)$  in the third quadrant. Clearly, the density associated with  $(X_0, Y_0)$  is equal to zero, while the density associated with  $(-X_0, -Y_0)$  is not. Had the distribution been symmetric, then the two points would have been characterized by the same probability mass. Hence, differences in densities between any point  $(x, y)$  and its symmetric counterpart  $(-x, -y)$  provide a measure of asymmetry for the joint distribution of  $X$  and  $Y$ . This simple figure illustrates the motivation for using equation (6) to test for symmetry.

### C. Entropy Measure of Asymmetry

The important question is how to test the null hypothesis of symmetry provided in (6). Intuitively, the joint distribution is symmetric if the distance between  $f^+(c)$  and  $f^-(c)$  is zero almost everywhere. Otherwise, it is asymmetric. To formalize this null hypothesis, we rely on measures of distance between the two probability density functions.

Originating in physics and information theory as a measure of uncertainty, entropy has a long history as a measure of divergence between distributions. It was first introduced by Shannon (1948) and later extended by Kullback and Leibler (1951).<sup>5</sup> More recently, entropy has drawn attention from financial economists. Recent examples include Cabrales, Gossner, and Serrano (2013), who use Shannon (1948)'s entropy to quantify the informativeness of ruin-averse investors' beliefs on the state of nature. Backus, Chernov, and Zin (2014) use Kullback and Leibler (1951)'s relative entropy to measure differences between physical and risk-neutral probabilities and derive appropriate bounds for the stochastic discount factors of consumption-based asset pricing models. In more recent work, Backus, Boyarchenko, and Chernov (2016) propose the use of co-entropy as a

---

<sup>5</sup>Ullah (1996) provides an excellent survey of various entropy measures and their wide applications in econometrics.

new measure of co-dependence between random variables; this is a bivariate variant of the relative entropy concept proposed in Backus, Chernov, and Zin (2014). Chabi-Yo and Colacito (2016) propose an alternative co-entropy measure to evaluate the performance of international finance models. In Section V, we present a comparison of our methodology to these co-entropy measures proposed in the literature.

The entropy measure we use belongs to the same  $K$ -class entropy as the Kullback-Leibler relative entropy. First proposed by Granger, Maasoumi, and Racine (2004), the measure is a special case of  $K$ -class entropy, with  $K = 1/2$ , which is a normalization of the Hellinger distance measure and the only metric entropy within its class. As shown in Granger, Maasoumi, and Racine (2004), this metric measure has many other desirable properties.

Consider first, for simplicity, the case in which we have only one exceedance level  $c$ . The entropy measure of asymmetry is defined as

$$(7) \quad S_{\rho}(c) = \frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (f^{+}(c, \tilde{x}, \tilde{y})^{\frac{1}{2}} - f^{-}(c, \tilde{x}, \tilde{y})^{\frac{1}{2}})^2 d\tilde{x} d\tilde{y},$$

which is a function of the exceedance level  $c$ . In practice,  $c$  is chosen according to empirical interests. For example, when  $c = 0$ ,  $S_{\rho}(0)$  measures the asymmetric comovement in the first quadrant (where both standardized returns are positive) and third quadrant (where both standardized returns are negative). If the asymmetric comovement in the tails of the distribution is of interest,  $S_{\rho}(c)$  can be measured at other exceedance levels, such as 0.5 or one standard deviations away from the mean.

The entropy measure  $S_{\rho}(c)$  is well defined for both continuous and discrete data. It takes values between zero and one and equals zero if and only if the densities are equal, which indicates symmetric comovement. Mathematically, it is a true measure of distance because it satisfies the triangular inequality. Moreover, the measure is invariant under continuous and strictly increasing

transformations, such as the common logarithm transformation.

Consider now the case in which we have multiple exceedance levels,  $c_1, \dots, c_m$ , and want to determine whether jointly symmetric comovement exists at all exceedance levels. For example, while the singleton set of  $c = \{0\}$  is usually of interest in the literature, the set of levels  $c = \{0, 0.5, 1, 1.5\}$  is also common in previous studies, such as Ang and Chen (2002) and Hong, Tu, and Zhou (2007). For the multi-level case, we can also apply the statistic in equation (7) to each of the individual levels and then aggregate the results. Since  $S_\rho(c)$  is a metric and is always non-negative, we could simply take the arithmetic average as an aggregation:

$$(8) \quad S_\rho = \frac{S_\rho(c_1) + \dots + S_\rho(c_m)}{m},$$

where  $S_\rho(c_j)$  is computed from equation (7) for  $j = 1, \dots, m$ . Thus, the entropy test statistic is well-defined for either the single test case with one exceedance level or the joint test case with multiple exceedance levels.

To carry out the entropy test in practice, we first need to estimate the joint density function of the data and then compute the integral in equation (7) to obtain the statistic  $\hat{S}_\rho(c)$ . Finally, we need to determine the distribution of the test statistic under the null hypothesis for computing the p-values of the test. This task is unfortunately much more complex than that of the asymmetric correlation test. The challenges inherent in the calculations are addressed in the following two subsections.

#### **D. Nonparametric Estimation**

In this subsection we provide a methodology to estimate the density functions in equation (7). Following Racine and Maasoumi (2007), among others, we use a non-parametric kernel smoothing method to consistently estimate unknown joint densities. Specifically, we use the popular Parzen-Rosenblatt kernel density estimator (Rosenblatt, 1956; Parzen, 1962). For the univariate case, the

Parzen-Rosenblatt kernel density estimator is defined as

$$(9) \quad \hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n k\left(\frac{X_i - x}{h}\right),$$

where  $n$  is the sample size of the data  $\{X_i\}$ ,  $h$  is a smoothing parameter (commonly referred to as the bandwidth), and  $k(\cdot)$  is a non-negative bounded kernel function. As we are dealing with bivariate density functions, in estimation we employ a commonly used product kernel function constructed as the product of univariate kernel functions. Our candidate kernel density estimator of the data is given by

$$(10) \quad \hat{f}(x, y) = \frac{1}{nh_1 h_2} \sum_{i=1}^n k\left(\frac{x_i - x}{h_1}\right) \times k\left(\frac{y_i - y}{h_2}\right),$$

where  $n$  is the sample size,  $k(\cdot)$  is a suitable univariate kernel function,  $h_1$  and  $h_2$  are the respective bandwidths for each of the two components, and  $\{(x_i, y_i)\}$  is an observed data pair. In the empirical applications below,  $n$  is set equal to  $T$ , the length of the return series.

The accuracy of the nonparametric kernel density estimator relies on the selection of both the kernel function and the bandwidth. Since the return data are continuous, we use a standard Gaussian kernel,  $k(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ , as the univariate kernel function in the estimation. The choice of kernel function is less important than the bandwidth selection. On selecting the bandwidth, we use the likelihood cross-validation method proposed by Duin (1976). This procedure solves the following maximization problem of the log-likelihood function;

$$(11) \quad \max_{h_1, h_2} \mathcal{L} = \sum_{i=1}^n \ln [\hat{f}_{-i}(x_i, y_i)],$$

where

$$(12) \quad \hat{f}_{-i}(x_i, y_i) = \frac{1}{nh_1 h_2} \sum_{j \neq i}^n k\left(\frac{x_j - x}{h_1}\right) \times k\left(\frac{y_j - y}{h_2}\right),$$

which is equal to  $\hat{f}(x, y)$  without the  $i$ th term. Econometrically, this data-driven bandwidth selection method yields density estimates that are close to the actual density in terms of the Kullback–Leibler relative entropy (for details, see Li and Racine, 2007).

Using the method described above, the density functions in equation (7) can be consistently estimated. The test statistic  $\hat{S}_\rho(c)$  can then be obtained by computing the integral using a standard numerical procedure in the case of a single exceedance level. In the presence of multiple exceedance levels, the test statistic is computed from equation (8).

## E. Distribution of the Test Statistic

To make statistical inferences for the entropy test of asymmetry, we need to know the sampling distribution of the test statistic under the null hypothesis of symmetry. There are two potential methods for determining the sampling distribution: asymptotic theory and the bootstrap resampling method.

The asymptotic theory for the class of entropies with similar functional forms was developed by Skaug and Tjøstheim (1993); Tjøstheim (1996) and Hong and White (2005). Under certain regularity conditions,  $\hat{S}_\rho(c)$  follows an asymptotically normal distribution and the distribution derived under the null hypothesis does not depend on the choice of bandwidth. This happens in part because the bandwidth is a quantity that vanishes in the limit. However, for a given finite sample size, the computed value of the test statistic critically depends on the bandwidth selection (see Maasoumi and Racine, 2008). This raises strong concerns about using the asymptotic distribution to conduct statistical inference in empirical applications. Therefore, following the suggestions of Hong and White (2005) and many others, rather than rely on an asymptotic distribution, we use the bootstrap resampling approach to determine the empirical null distribution of  $\hat{S}_\rho(c)$  (for further discussions on the bootstrap resampling approach, see Efron, 1982; Hall, 1992; Horowitz, 2001).<sup>6</sup>

---

<sup>6</sup>See Section IA.I of the internet appendix for a discussion of the choice of bootstrap resampling methods suitable

### III. Size and Power of the Entropy Test

In this section, using Copula-GARCH based Monte Carlo simulations, we examine the size and power of the entropy test and show that it has reliable size and fairly high power in finite samples.

#### A. Modeling Dependence with Copulas

Since we are testing the joint distribution of two random variables, the simulation requires generating random samples from a joint distribution with a certain dependence structure. Copula is a widely used method to model the complete dependence structure between random variables.

As documented in the literature (see, e.g., Ang and Chen, 2002; Ang, Chen, and Xing, 2006), stock returns usually show stronger downside comovement than the upside comovement with the market return. A Clayton copula that features stronger left tail dependence is a natural candidate for the data generating process (DGP). It has the functional form given by

$$C_{clay}(u, v; \tau) = (u^{-\tau} + v^{-\tau} - 1)^{-\frac{1}{\tau}},$$

where  $\tau > 0$  governs the dependence between the marginal distributions. A higher  $\tau$  indicates stronger left tail dependence.

A bivariate Gaussian copula features symmetric dependence and is defined as

$$C_{nor}(u, v; \rho) = \Phi_{\rho}(\Phi^{-1}(u), \Phi^{-1}(v)),$$

where  $\rho$  is the correlation coefficient between the marginal distributions,  $\Phi^{-1}$  is the inverse of standard normal cumulative distribution function, and  $\Phi_{\rho}$  is the standard bivariate normal distribution function with correlation  $\rho$ . To best fit our return data, we maximize flexibility by

---

for the entropy test of asymmetry and detailed procedures for implementing the resampling method.

using a mixed Gaussian–Clayton copula as the DGP,<sup>7</sup> which is defined as

$$(13) \quad C_{mix}(u, v; \rho, \tau, \kappa) = \kappa C_{nor}(u, v; \rho) + (1 - \kappa) C_{clay}(u, v; \tau),$$

where  $\kappa$  indicates the weight on the bivariate normal (Gaussian) copula. The mixed copula nests both the Gaussian and Clayton copulas as special cases: When  $\kappa = 1$ , the mixed copula reduces to a Gaussian copula; When  $\kappa = 0$ , the mixed copula reduces to a Clayton copula. In the simulations below, we use  $\kappa$  values of 0, 0.25, 0.375, 0.5, 0.75 and 1 to generate random samples with different levels of asymmetric comovement.

[Insert Figure 2 about here]

Figure 2 shows the contour plots of random samples simulated using Copula–GARCH models, which are described in detail in the following section. The model parameters are fitted to the value-weighted monthly returns of the smallest size portfolio and the market portfolio using the maximum likelihood method. The samples are generated by assuming Clayton, Gaussian and equally weighted mixed Gaussian–Clayton copula dependence structures, respectively. The contour plots of simulated data (sub-figures a, b and c) are plotted along with the actual return data (sub-figure d). It is clear that the Clayton copula generates data with a stronger lower tail dependence, since the plots are highly concentrated in the lower left tail. The smallest size portfolio is known to have the strongest asymmetric correlation, but the data plots generated by the pure Clayton copula seem to have stronger lower tail dependence than the actual data. As shown in sub-figure (c), the scatter plots generated by the equally weighted mixed Gaussian–Clayton copula look most similar to the actual data plots in panel (d), which justifies our decision to use mixed copulas as the DGP in the simulations.

---

<sup>7</sup>Tawn (1988) proves that every convex combination of existing copula functions is still a copula.

## B. Simulation with Copula–GARCH Models

To ensure that the random samples drawn from the mixed Gaussian–Clayton copula model are as close to the actual data as possible, we must specify a model that best mimics the marginal distribution of the asset returns. We model an asset return as the sum of an expected return component and a residual following a GARCH(1,1) process, a parsimonious way to model conditional heteroskedasticity. In the implementation, we first fit the data to the copula–GARCH model using the maximum likelihood method. We then plug the maximum likelihood estimates back into the model and use them as the true parameters to generate data to compute the simulated p-values. To be conservative, instead of using portfolios that clearly show asymmetric comovement, such as the smallest size or momentum portfolios, we use the value-weighted returns of the fourth largest size portfolio<sup>8</sup> and the market portfolio to estimate the copula and GARCH parameters. The maximum likelihood estimates of the Clayton copula parameter  $\tau$  and Gaussian copula parameter  $\rho$  are 5.768 and 0.951, respectively. The estimated GARCH parameters are reported in Table IA.1 of the internet appendix. The results show that all the GARCH parameters are statistically significant at the 5% level, indicating that the GARCH(1,1) is indeed a good description of the return data. It is of interest to see whether the entropy test has reasonable power under the parameter settings.<sup>9</sup>

In summary, we simulate the data as follows:

1. For a given  $\kappa$ , draw a bivariate uniform random sample of size  $T$  from the mixed Gaussian–Clayton copula model;

---

<sup>8</sup>Based on the entropy test results in Section IV, the fourth largest size portfolio is the smallest size portfolio that cannot reject the null of symmetry at the 5% significance level.

<sup>9</sup>We also check robustness of the test power using Threshold GARCH(1,1) model as the underlying DGP for the marginal distributions. The estimated TGARCH parameters are reported in Table IA.2 of the internet appendix. The test results, reported in Tables IA.5, IA.6, and IA.7 in the internet appendix, are qualitatively similar. The asymmetry of the joint distribution is mainly governed by the copula function. Therefore, the leverage effect in the conditional heteroskedasticity of the marginal distributions has little effect on the joint asymmetry.

2. Apply an inverse standard normal cumulative distribution function transformation to obtain a bivariate standard normal random sample with a pre-specified dependence structure;
3. Feed each series of the joint normal random sample into the univariate GARCH(1,1) process as the innovation terms to generate a simulated joint return series;<sup>10</sup>
4. Repeat steps 1 to 3 for 1,000 times;
5. Repeat steps 1 to 4 with different sample sizes  $T = \{240, 420, 600, 840\}$ .

Our choice of sample sizes represents typical data lengths encountered in empirical studies:  $T = 240$  represents 20 years of monthly frequency data,  $T = 420$  is the length of the subsample data period used in Hong, Tu, and Zhou (2007),  $T = 600$  represents 50 years of monthly frequency data and is close to the full sample data length ( $T = 588$ ) used in this paper, and  $T = 840$  represents 70 years of monthly frequency data. In the simulation, we use one fixed bandwidth for each 1,000 random samples generated from the same DGP. The fixed bandwidth is set to the average of the 1,000 bandwidths computed for each of the 1,000 random samples via the likelihood cross-validation. The optimal block length selection is similar in that the expected block length for each 1,000 random samples generated from the same DGP is fixed to be the average of the 1,000 optimal block lengths computed using Patton, Politis, and White (2009)'s algorithm. Averaging the bandwidths and block lengths across random samples drawn from the same DGP can potentially reduce sampling randomness and make the simulation results more stable.<sup>11</sup>

[Insert Table 1 about here]

Table 1 reports the empirical size and power for both tests based on 1,000 Monte Carlo

---

<sup>10</sup>It ensures that the simulated data vectors each follow a GARCH(1,1) process and the desired dependence structure is governed by the mixed copula model.

<sup>11</sup>We also use the optimal bandwidth and block length for each simulated sample without averaging and obtain similar results.

simulations in which the nominal size is set to 5%. Powers are reported for different DGPs with different asymmetric comovement levels (from  $\kappa = 1$  to  $\kappa = 0$ ) and at various sample sizes. Following Hong, Tu, and Zhou (2007), we focus on exceedance levels  $\{0\}$  and  $\{0, 0.5, 1, 1.5\}$ . For comparison, we also report the size and power of the HTZ test. When the underlying DGP is symmetrically distributed with  $\kappa = 1$ , the true probability of rejection is the size of the tests. The empirical sizes of both tests are reported in Panel A of Table 1. The empirical size of the entropy test at  $\{c = 0\}$  is fairly close to 5% and approaches the true nominal size as the sample size increases. When the sample size is 840, the empirical size is 0.055, which is very close to 5%, the nominal size. The joint entropy test at  $c = \{0, 0.5, 1, 1.5\}$  has an empirical size of 0.047 with a sample size of 840, which is also close to the nominal size. We find that the HTZ test shows a tendency toward under-rejection at both the single and joint exceedance levels, with empirical sizes too close to zero in many cases. Similar bootstrap procedures can improve the HTZ test so that its size is as reliable as that of the entropy test, but its power, on average, remains comparatively low.

We next evaluate the empirical powers of both tests. We find that, in most cases, the single test has higher power than the joint test does; this is the case for both HTZ and entropy tests. The results indicate that the entropy test, in most cases, has higher average rejection rates than the HTZ test at both the single and joint exceedance levels. The difference in power varies with the dependence structure of the DGP. When the simulated data are strongly asymmetric, both tests perform similarly. If the DGP is a bivariate Clayton copula ( $\kappa = 0$ ) with an exceedance level of  $c = 0$ , the difference in power is not large (0.211 for  $T = 240$ ) and vanishes as the sample size increases to  $T = 840$ . However, the power difference becomes most pronounced when the degree of asymmetry is not very strong. For example, when the DGP is a 50% mixed Gaussian-Clayton copula, the power of the entropy test is about twice as high as the power of the HTZ test, even when the sample size is 840. In general, both tests increase in power as the sample size increases.

Our results are qualitatively similar to those in Table 1 if the nominal size is set at 1% or

10% (see Tables IA.3 and IA.4 of the internet appendix for details). This fact reaffirms that the proposed entropy test generally has the correct size and higher power than the HTZ test. There are two possible reasons for this. First, the entropy measure utilizes more information from the joint distribution, giving it an information advantage over the HTZ test, which uses only the second moment. Second, we rely on the bootstrap resampling method for inference, so the null distribution of the test statistics is potentially more accurate than the asymptotic distribution.

## **IV. Is Asymmetry Rare?**

In this section, we apply our entropy measure to test whether statistically significant asymmetry exists in common portfolios sorted by size, book-to-market ratio, and momentum. Previous studies have rarely found asymmetry; however, as we show below, the entropy test shows very different results.

### **A. Test Portfolios**

Following prior studies on asymmetric correlation, we consider portfolios of stocks sorted by market capitalization, book-to-market ratio, and momentum. We use value-weighted returns for all portfolios. The return on the Center for Research in Security Prices (CRSP) value-weighted market index, based on stocks listed on the NYSE/AMEX/NASDAQ, is used as a proxy for the market return. All returns are calculated at a monthly frequency and in excess of the risk-free rate, which is defined as the one-month T-bill rate. All of these data are available on Kenneth French's website.<sup>12</sup> The sample period is from January 1965 to December 2013 (588 observations in total).<sup>13</sup>

---

<sup>12</sup>We are grateful to Kenneth French for making the data available at [http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

<sup>13</sup>We choose the starting date to be consistent with Hong, Tu, and Zhou (2007). The results are similar when using January 1963 as the starting date. We also report the test results using exactly the same sample period as in Hong, Tu, and Zhou (2007) (January 1965 to December 1999) in Table IA.8 of the internet appendix.

## B. Test Results

Table 2 shows the asymmetry test results based on both the HTZ test and the entropy test with the exceedance levels of  $\{0\}$  and  $\{0, 0.5, 1, 1.5\}$ . Panel A of Table 2 provides the test results for the size portfolios. At the usual 5% level, the entropy test at the single exceedance level  $c = \{0\}$  rejects symmetry for the six smallest size portfolios. We find even more significant asymmetry when testing at the multiple exceedance levels  $c = \{0, 0.5, 1, 1.5\}$ . Since the joint test puts more weight on densities in the tails, this finding indicates that asymmetric comovement is more prominent in the tail regions of the joint distribution. In contrast, the HTZ test can only reject symmetry for the smallest size portfolio, regardless of whether we use single or multiple exceedance levels. It is interesting to note that the entropy test statistics generally decrease as firm size increases; a similar pattern also holds for the HTZ test statistics. This finding is consistent with the work of Ang and Chen (2002).

[Insert Table 2 about here]

For comparison, the last two columns of Panel A of Table 2 also report conventional moment-based statistics that describe asymmetry: the skewness and coskewness. The skewness does not show any pattern across size portfolios, while the coskewness is generally larger in absolute value for smaller stocks. Clearly, these moment-based measures alone do not provide enough information to draw any conclusions regarding the relationship between asymmetry in stock returns and firm size. In contrast, our entropy test provides evidence of strong asymmetry for the six smallest size portfolios.

Panel B of Table 2 reports the test results for book-to-market portfolios. Unlike the size portfolios, there are no apparent patterns for the entropy statistic across stocks with different book-to-market ratios. The same holds true for the HTZ test statistic, skewness, and coskewness. At the 10% level, the entropy test at  $c = \{0\}$  rejects seven out of 10 book-to-market portfolios and the

joint entropy test rejects nine portfolios, while the HTZ tests reject none.

Panel C of Table 2 reports the test results for the momentum portfolios. The single entropy test finds statistically significant asymmetry in the returns of both the past loser and winner portfolios at the 1% level. The joint version of the entropy test detects significant asymmetry in the top five winner portfolios at the conventional 5% significance level. Significant asymmetry in the past loser portfolio is consistent with the results of Ang and Chen (2002), who show that the bivariate normal model is rejected when fitted to the past loser portfolio returns; in other words, the returns of the past loser portfolios exhibit asymmetric correlations. In contrast, the HTZ test detects hardly any significant asymmetric correlation among the momentum portfolios at the 5% level. Only the past loser portfolio shows marginal significance at the 10% level, with a p-value of 0.098. We find that the entropy test statistic  $S_p$  is largest at the lower and higher ends and smaller in the middle deciles. The  $J_p$  statistic in the HTZ test follows a similar pattern. Finally, no obvious pattern of asymmetry can be inferred from skewness or coskewness.

## **V. Asset Pricing and Asymmetric Comovement**

What are the asset pricing implications of the asymmetric comovement of individual stocks with the market return that we found in many portfolios? In this section, we introduce a downside asymmetric comovement measure based on the entropy test statistic and examine its effect on the cross-sectional variations of expected returns.

### **A. Downside Asymmetric Comovement**

We have shown that the entropy test statistic  $S_p(c)$  defined in equation (7) can successfully capture the degree of asymmetric comovement in the stock market, since the entropy test demonstrates good finite sample power.  $S_p(c)$  is a normalized metric that is always non-negative and therefore does not indicate whether stock comovement with the market is stronger in the downside or upside.

However, investors are generally more concerned about downside risk and require higher premiums for holding assets with higher downside risk. Similar to uncertainty in Segal, Shaliastovich, and Yaron (2015), we also need a measure that distinguishes the direction of asymmetry.

We define a lower quadrant probability (LQP) and an upper quadrant probability (UQP) as follows:

$$(14) \quad \text{LQP} = \Pr(\tilde{x} \leq -c, \tilde{y} \leq -c) = \int_{-\infty}^{-c} \int_{-\infty}^{-c} f(\tilde{x}, \tilde{y}) d\tilde{x}d\tilde{y},$$

$$(15) \quad \text{UQP} = \Pr(\tilde{x} \geq c, \tilde{y} \geq c) = \int_c^{+\infty} \int_c^{+\infty} f(\tilde{x}, \tilde{y}) d\tilde{x}d\tilde{y}.$$

These variables measure the concentration of individual stock and market return pairs being either above or below the exceedance level  $c$ . When  $c = 0$ , a higher LQP (UQP) indicates a higher tendency for the stock to comove with the market below (above) average levels, thus capturing the downside (upside) comovement of individual stock returns with the market. We define our downside asymmetric comovement (DownAsy) as

$$(16) \quad \text{DownAsy} = \text{Sign}(\text{LQP} - \text{UQP})S_\rho(c),$$

where  $\text{Sign}(\cdot)$  is a sign function that takes the value of 1 if the difference between LQP and UQP is non-negative and  $-1$  otherwise. Higher DownAsy indicates greater downside risk relative to upside potential.

To see the intuition behind DownAsy, consider the case where the stochastic discount factor (SDF) of an asset pricing model is only a function of the market return that is given by  $g(R_{m,t+1})$ , of which the capital asset pricing model (CAPM) is a special case. Let  $p_t$  be the price of an asset today, and  $x_{t+1}$  be its future payoff in the next period. Then  $p_t$  is the discounted expected payoff,

$$p_t = E_t[g(R_{m,t+1})x_{t+1}].$$

We assume that the SDF is positive to eliminate arbitrage (Hansen and Jagannathan, 1991). It is then clear that  $g(R_{m,t+1})x_{t+1}$  is negative in the third quadrant. Holding the properties of the asset returns constant outside the exceedance regions, then the more density mass in the third quadrant, the lower the current price. If the mass of the density in the third quadrant is greater than the mass in the first quadrant, then the asset return is asymmetric to the downside. In this case, DownAsy lowers the current price implying in general a risk premium for the downside asymmetry risk. For example, for a class of utility functions, Gul (1991) and Ang, Bekaert, and Liu (2005) rationalize that investors weigh payoffs below a certain reference point more heavily than those above it. This suggests that the downside asymmetry risk premium from their models can be higher than that from a typical model with CRRA utility.

## **B. Downside Asymmetric Comovement and Firm Characteristics**

We construct the comovement measures LQP, UQP, and DownAsy, as well as downside beta  $\beta^-$ , upside beta  $\beta^+$ , coskewness, and cokurtosis based on CRSP data from January 1962 to December 2013. The appendix provides detailed formulas for these variables. The estimation is carried out using realized daily return data from overlapping 12-month periods. We drop any year with fewer than 100 daily return observations for each stock. The estimates are then updated monthly.

[Insert Table 3 about here]

Table 3 reports the time series averages of the cross-sectional correlations of the lower quadrant probability, the upper quadrant probability, downside asymmetric comovement (DownAsy), the CAPM beta ( $\beta$ ), downside beta ( $\beta^-$ ), upside beta ( $\beta^+$ ), the log of market capitalization (Size), the log of the book-to-market ratio (Bm), the turnover ratio (Turn), the normalized Amihud illiquidity measure (Illiq), the past six-month return (Mom), idiosyncratic volatility (Ivol), coskewness (Coskew), cokurtosis (Cokurt), and the maximum daily return over the past month (Max). It is

interesting to note that the correlation between LQP and UQP is fairly modest ( $-0.080$ ), which means that the tendencies of the stock to comove with market in the upside or downside are almost linearly independent. This finding justify our approach to separately estimate the lower and upper quadrant probabilities.

Both LQP and UQP are positively correlated with the CAPM beta with correlation coefficients of 0.463 and 0.436, respectively.  $\beta$  captures the linear dependence between an individual stock return and the market return, while quadrant probabilities measure the tendency of the returns to occur above (below) the sample averages. Stocks with a higher  $\beta$  will have a higher probability of being above (below) their sample means when the market is above (below) average. LQP and UQP have little correlation with coskewness. However, they have high positive correlations with the cokurtosis, since cokurtosis measures the fatness of the tails in a given joint distribution and a fatter tail indicates a higher probability in that quadrant. DownAsy has little correlation with firm characteristics such as size, book-to-market ratio, etc., which indicates that DownAsy is distinct from these well-known measures and provides additional new information.

In Table 4 we sort stocks into deciles based on their realized DownAsy values each month. We report the time series averages, which range from  $-0.065$  to  $0.107$ , in the second column. None of the other firm characteristics display a similar monotonic pattern. This result reaffirms the findings in Table 3 that DownAsy is not similar to other firm characteristics. Therefore, if DownAsy explains the cross-section of asset returns, then its impact on asset pricing is different from the effects of these known firm characteristics documented in the literature.

[Insert Table 4 about here]

### C. Comparison with other asymmetry measures

We sort stocks into quintiles based on their realized asymmetry measures estimated over the previous 12 months. We then compute the equal-weighted average returns on these portfolios. Since we need the first 12 months of data to estimate the measures, the quintile portfolio returns are calculated from January 1963 to December 2013.

[Insert Table 5 about here]

Panel A of Table 5 reports the univariate portfolio sorting results based on the DownAsy and other conventional asymmetry measures. The average returns of the DownAsy portfolios increase from 0.65% to 1.03% per month. The trading strategy of investing in high-DownAsy stocks and shorting low-DownAsy stocks therefore yields an economically significant one-month return of 0.38%, which amounts to a return premium of 4.5% per annum. The return difference is also statistically significant at the 1% level. To attribute this excess return to exposure to previously identified systematic risk factors, we regress the return series of each quintile portfolio and the spread portfolio on Carhart (1997)'s four factors. The abnormal return of the spread portfolio after adjusting for the Carhart four factors is 0.19% per month (2.3% per annum). It is statistically significant at the 5% level and economically meaningful.

For comparison, we examine the asset pricing implications of other conventional asymmetry measures. Consider, first, an asymmetric correlation measure based on the HTZ test statistic. Since the test statistic is always positive and does not indicate whether an exceedance correlation exists in the downside or in the upside, we need to create a modified version similar to DownAsy. We define the downside asymmetric correlation, denoted DownCorr, as

$$(17) \quad \text{DownCorr} = \text{Sign}(\rho^-(c) - \rho^+(c))J_\rho(c),$$

where  $c$  is the exceedance level. DownCorr takes the value of  $J_\rho(c)$  if  $\rho^-(c) - \rho^+(c) \geq 0$  — that is, the exceedance correlation in the downside tail is greater than or equal to that in the upside tail — and takes the value of  $-J_\rho(c)$  otherwise.

We report the results for the quintile portfolios sorted by DownCorr ( $c = 0$ ). Similar to DownAsy, the greater the downside asymmetric correlation, the greater the expected return. However, the return difference is lower — 0.24% per month or 2.9% per annum, which is about 36% less than the spread return of DownAsy portfolios. Moreover, this return difference can be fully explained by the Carhart four factors. Similarly, although there are average return differences between stocks in the highest and lowest downside beta (Ang, Chen, and Xing, 2006) and coskewness (Harvey and Siddique, 2000) quintiles, these differences can be fully explained by the Carhart four factors.

Finally, we compare DownAsy with other downside asymmetry measures based on the co-entropies proposed by Backus, Boyarchenko, and Chernov (2016) and Chabi-Yo and Colacito (2016). This is an important robustness check. If there is a genuine asymmetry risk premium, these measures should detect that premium as well. To apply these alternative co-entropy measures to our data, we need to first compute exceedance co-entropies, which are simply the co-entropies evaluated in the exceedance regions. Specifically, based on the co-entropy in Backus, Boyarchenko, and Chernov (2016), we compute the exceedance co-entropies as

$$(18) \quad C_{BBC}^+(c, x, y) = \frac{L(xy) - L(x) - L(y)}{L(x) + L(y)} \Big|_{(x > c, y > c)},$$

$$(19) \quad C_{BBC}^-(c, x, y) = \frac{L(xy) - L(x) - L(y)}{L(x) + L(y)} \Big|_{(x < -c, y < -c)};$$

and, based on the definition of the co-entropy given in Chabi-Yo and Colacito (2016), we compute

another set of exceedance co-entropies as

$$(20) \quad C_{CC}^+(c, x, y) = \frac{L(x) + L(y) - L(x/y)}{L(x) + L(y)} \Big|_{(x > c, y > c)},$$

$$(21) \quad C_{CC}^-(c, x, y) = \frac{L(x) + L(y) - L(x/y)}{L(x) + L(y)} \Big|_{(x < -c, y < -c)},$$

where  $L(x) = \log E(x) - E(\log x)$ . Then,  $C_{BBC}^-(c, x, y) - C_{BBC}^+(c, x, y)$  and  $C_{CC}^-(c, x, y) - C_{CC}^+(c, x, y)$  can both serve as alternative downside asymmetry measures. These two measures are computationally much easier to implement than DownAsy since they are defined based on expectations that can be consistently estimated by sample means.

Panel B of Table 5 reports the portfolio returns when stocks are sorted by the downside asymmetry measures based on the co-entropies of Backus, Boyarchenko, and Chernov (2016) and Chabi-Yo and Colacito (2016), respectively, which are referred to as BBC and CC in the table. Despite apparent differences in these co-entropy measures, both the BBC and CC measures are equally successful in measuring asymmetry. The return spreads between the highest-asymmetry and lowest-asymmetry portfolios are approximately 0.27% per month (3.24% per annum) in both cases, higher than the return spread generated by DownCorr. This indicates, as expected theoretically, that the co-entropy-based asymmetry measures capture the stocks' asymmetric comovement better than the correlation-based measure, although these spreads are still lower than that of the portfolios formed using DownAsy. This is also unsurprising, as DownAsy utilizes all the information available in the entire joint return distribution while neither of the co-entropy measures does. However, if we take the estimation error into account, the 95% confidence interval of the return spread for portfolios sorted by the BBC measure is approximately [0.148, 0.391]. Statistically, the asymmetry risk premiums of DownAsy and the co-entropy-based downside asymmetry measures could possibly be equally high. This suggests that the co-entropy measures are good alternatives for DownAsy in this context. Overall, our finding of a substantial downside

risk premium is robust, as all the entropy-based measures yield statistically significant evidence.

#### **D. Further Analysis**

We next examine how asymmetry relates to investor sentiment, market volatility and market liquidity. In addition, we determine whether our asymmetry measures remain statistically significant after controlling for asymmetric correlation, asymmetric beta or coskewness via a double sorting procedure.

The average return differences between stocks with high and low DownAsy change over time, with positive realizations in about 62% of the time periods. Columns (1) to (4) of Table 6 report the results of regressing the return difference on market volatility, investor sentiment (Baker and Wurgler, 2006) and market liquidity (Pástor and Stambaugh, 2003).<sup>14</sup> The asymmetry risk premium is strong during periods of low investor sentiment, low market volatility and high market liquidity, though the relation to investor sentiment is not statistically significant. We also estimate DownAsy for each quarter, then sort all stocks into quintiles according to their average quarterly asymmetry during the previous 12 months. Columns (5) to (8) of Table 6 report the regression results with the return difference as the dependent variable, using the average quarterly asymmetry to sort portfolios. We can see that the findings are similar, except that the negative relation to investor sentiment becomes statistically significant.

[Insert Table 6 about here]

In order to see the incremental effect of DownAsy on stock returns after controlling for other asymmetry measures, we first sort stocks into quintiles by a given asymmetry measure, such as DownCorr,  $\beta -$ , or coskewness. Within each quintile, we further divide stocks by DownAsy. The results are reported in Table 7.

---

<sup>14</sup>They are available at <http://people.stern.nyu.edu/jwurgler/> and <http://faculty.chicagobooth.edu/lubos.pastor/research/>.

[Insert Table 7 about here]

Panel A reports the results from sorting first by DownCorr and then by DownAsy. The return difference remains sizable after controlling for downside asymmetric correlations, ranging from 0.282% to 0.501% across the five downside correlation groups. All are statistically significant at the 5% level and economically significant. The results indicate that the downside asymmetric correlation cannot rule out the importance of DownAsy. Panel B of Table 7 reports the results based on  $\beta^-$ . Again, after controlling for the effect of the downside beta, economically and statistically significant return differences persist. Similar results can be seen in Panel C for coskewness.

Overall, we confirm that our entropy-based downside asymmetry measure contains information that is not captured by other asymmetry measures. Therefore, it offers a new perspective on asymmetry for practical use, complementing existing measures.

## VI. Conclusion

Asymmetric comovement in stock returns is important for both portfolio management and risk hedging. However, existing measures of asymmetry use only the second moments — a special case of joint asymmetry. In particular, HTZ's asymmetric correlation test only measures linear dependence, ignoring higher-order dependence. Therefore, it has low power and can only detect a few cases of asymmetry for commonly used stock portfolios. In this paper, we propose a model-free entropy measure that provides a direct test for asymmetry in the joint distribution. Econometrically, our test extends the univariate asymmetry test proposed by Racine and Maasoumi (2007) to the bivariate case, which has important applications in finance. Using Monte Carlo simulations, we find that the entropy test has reliable size and good power. Empirically, based on the entropy test, we find statistically significant asymmetry for many common portfolios, such as size, book-to-market, and momentum portfolios.

Our paper also examines whether stocks' downside asymmetric comovement with the market has pricing implications in the cross-section of stock returns. Using a modified entropy measure based on quadrant probabilities of the joint distribution of stock and market returns, we find a significant return difference between stocks with high and low downside asymmetry. The findings are consistent with the theoretical implications of a representative agent model with disappointment aversion utility. A trading strategy that buys stocks with high downside asymmetry and sells stocks with low downside asymmetry can earn a return of 4.5% per year. Such a premium is both economically and statistically significant and cannot be explained by previously identified risk factors, such as the excess return on the market, SMB (Small Minus Big), HML (High Minus Low), or momentum factors. Further analysis indicates that the premium also cannot be explained by other moment-based measures, such as the downside beta and coskewness.

## Appendix: Variable Definitions

In this appendix, we provide detailed definitions for various variables used in the paper. A stock  $i$ 's demeaned daily excess return is denoted by  $\tilde{r}_{i,d}$ , and the demeaned daily market excess return is denoted by  $\tilde{r}_{m,d}$ .

CAPM BETA:  $\beta$  is estimated for each month  $t$  using past 12-month observations, based on the standard formula

$$(A-1) \quad \hat{\beta}_{i,t} = \frac{\sum_{d=1}^{d=D_t} \tilde{r}_{i,d} \tilde{r}_{m,d}}{\sum_{d=1}^{d=D_t} \tilde{r}_{m,d}^2},$$

where  $D_t$  is the number of trading days in the past 12-month period ending in month  $t$ .

DOWNSIDE and UPSIDE BETAS: We denote the sample average of the daily market excess return during a 12-month period ending in month  $t$ ,  $\hat{\mu}_{m,t}$ . We further denote the demeaned excess return and demeaned market excess return conditional on market excess return being below (above)  $\hat{\mu}_{m,t}$  as  $\tilde{r}_{i,d}^-$  ( $\tilde{r}_{i,d}^+$ ) and  $\tilde{r}_{m,d}^-$  ( $\tilde{r}_{m,d}^+$ ), respectively. Following the definitions in Ang, Chen, and Xing (2006), we compute downside and upside betas as

$$(A-2) \quad \hat{\beta}_{i,t}^- = \frac{\sum_{r_{m,d} < \hat{\mu}_{m,t}} \tilde{r}_{i,d}^- \tilde{r}_{m,d}^-}{\sum_{r_{m,d} < \hat{\mu}_{m,t}} \tilde{r}_{m,d}^{-2}}, \text{ and } \hat{\beta}_{i,t}^+ = \frac{\sum_{r_{m,d} > \hat{\mu}_{m,t}} \tilde{r}_{i,d}^+ \tilde{r}_{m,d}^+}{\sum_{r_{m,d} > \hat{\mu}_{m,t}} \tilde{r}_{m,d}^{+2}}.$$

COSKEWNESS (Coskew): Following Harvey and Siddique (2000), the coskewness of stock  $i$  over a 12-month period ending in month  $t$  is given by

$$(A-3) \quad \widehat{\text{coskew}}_{i,t} = \frac{\frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{i,d} \tilde{r}_{m,d}^2}{\sqrt{\frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{i,d}^2 \left( \frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{m,d}^2 \right)}},$$

where  $D_t$  is the number of trading days in a 12-month period ending in month  $t$ .

COKURTOSIS (Cokurt): The cokurtosis of stock  $i$  over a 12-month period ending in month  $t$

is similarly defined as

$$(A-4) \quad \widehat{\text{cokurt}}_{i,t} = \frac{\frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{i,d} \tilde{r}_{m,d}^3}{\sqrt{\frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{i,d}^2 \left( \frac{1}{T} \sum_{d=1}^{d=D_t} \tilde{r}_{m,d}^2 \right)^{3/2}}},$$

where  $D_t$  is the number of trading days in a 12-month period ending in month  $t$ .

**IDIOSYNCRATIC VOLATILITY (Ivol):** The measure Ivol of stock  $i$  at the end of each month  $t$  is defined as the standard deviation of the CAPM residual series over the past 12 months.

**SIZE:** Firm size for each month  $t$  is measured using the natural logarithm of the market value of equity at the end of month  $t$ .

**BOOK-TO-MARKET RATIO (Bm):** Following Fama and French (1992), a firm's book-to-market ratio is calculated using its market value of equity and book value of equity. We assume the book value is available six months after the reporting date. We take book values from COMPUSTAT supplemented by book values from Ken French's website. The Bm at month  $t$  is defined as the natural logarithm of the book-to-market ratio at the end of month  $t$ .

**MOMENTUM (Mom):** Following Jegadeesh and Titman (1993), the momentum effect of each stock in month  $t$  is measured by the cumulative return over the previous six months, with one month skipped; i.e., the cumulative return from month  $t - 6$  to month  $t - 1$ .

**TURNOVER (Turn):** The turnover ratio is calculated monthly as the monthly trading volume divided by month-end shares outstanding.

**ILLIQUIDITY (Illiq):** We first calculate the ratio of absolute price change to dollar trading volume for each stock each day. Then we take the monthly average of the ratio if the number of observations is greater than 15 in that month. Following Acharya and Pedersen (2005), we normalize the Amihud ratio and truncate it at 30.

**MAXIMUM (Max):** Following Bali, Cakici, and Whitelaw (2011), Max for stock  $i$  in month  $t$  is defined as the maximum daily excess return that month.

## References

- Acharya, V. V., and L. H. Pedersen. 2005. Asset pricing with liquidity risk. *Journal of Financial Economics* 77:375–410.
- Amaya, D., P. Christoffersen, K. Jacobs, and A. Vasquez. 2015. Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics* 118:135–67.
- Ang, A., G. Bekaert, and J. Liu. 2005. Why stocks may disappoint. *Journal of Financial Economics* 76:471–508.
- Ang, A., and J. Chen. 2002. Asymmetric correlations of equity portfolios. *Journal of Financial Economics* 63:443–94.
- Ang, A., J. Chen, and Y. Xing. 2006. Downside risk. *Review of Financial Studies* 19:1191–239.
- Backus, D., N. Boyarchenko, and M. Chernov. 2016. Term structures of asset prices and returns. Working Paper, National Bureau of Economic Research.
- Backus, D., M. Chernov, and S. Zin. 2014. Sources of entropy in representative agent models. *The Journal of Finance* 69:51–99.
- Bae, K.-H., G. A. Karolyi, and R. M. Stulz. 2003. A new approach to measuring financial contagion. *Review of Financial studies* 16:717–63.
- Baker, M., and J. Wurgler. 2006. Investor sentiment and the cross-section of stock returns. *The Journal of Finance* 61:1645–80.
- Bali, T. G., N. Cakici, and R. F. Whitelaw. 2011. Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics* 99:427–46.
- Ball, R., and S. Kothari. 1989. Nonstationary expected returns: Implications for tests of market efficiency and serial correlation in returns. *Journal of Financial Economics* 25:51–74.

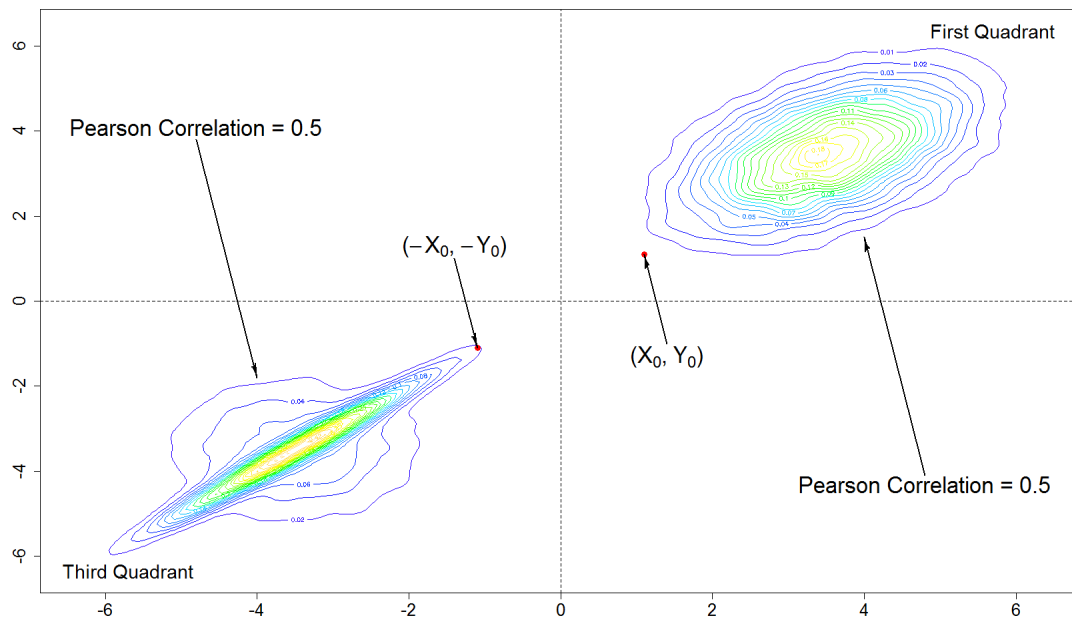
- Bekaert, G., and E. Engstrom. 2016. Asset return dynamics under habits and bad-environment good-environment fundamentals. *Journal of Political Economy*, forthcoming.
- Bekaert, G., E. Engstrom, and A. Ermolov. 2015. Bad environments, good environments: A non-gaussian asymmetric volatility model. *Journal of Econometrics* 186:258–75.
- Bekaert, G., and G. Wu. 2000. Asymmetric volatility and risk in equity markets. *Review of Financial Studies* 13:1–42.
- Cabrales, A., O. Gossner, and R. Serrano. 2013. Entropy and the value of information for investors. *The American Economic Review* 103:360–77.
- Carhart, M. M. 1997. On persistence in mutual fund performance. *The Journal of finance* 52:57–82.
- Chabi-Yo, F., and R. Colacito. 2016. The term structures of co-entropy in international financial markets. Available at SSRN: <https://ssrn.com/abstract=2341772> .
- Chabi-Yo, F., S. Ruenzi, and F. Weigert. 2014. Crash sensitivity and the cross-section of expected stock returns. Available at SSRN 2011746 .
- Cho, Y.-H., and R. F. Engle. 1999. Time-varying betas and asymmetric effect of news: Empirical analysis of blue chip stocks. Working Paper, National Bureau of Economic Research.
- Colacito, R., E. Ghysels, J. Meng, and W. Siwasarit. 2016. Skewness in expected macro fundamentals and the predictability of equity returns: Evidence and theory. *Review of Financial Studies* 29:2069–109.
- Conrad, J., R. F. Dittmar, and E. Ghysels. 2013. Ex ante skewness and expected stock returns. *The Journal of Finance* 68:85–124.
- Conrad, J., M. N. Gultekin, and G. Kaul. 1991. Asymmetric predictability of conditional variances. *Review of Financial Studies* 4:597–622.

- Cont, R. 2001. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance* 1:223–36.
- Dahlquist, M., A. Farago, and R. Tédongap. 2017. Asymmetries and portfolio choice. *Review of Financial Studies* 30:667–702.
- Duin, R. P. 1976. On the choice of smoothing parameters for parzen estimators of probability density functions. *IEEE Transactions on Computers* .
- Efron, B. 1982. *The jackknife, the bootstrap, and other resampling plans*, vol. 38. Society for industrial and applied mathematics.
- Fama, E. F., and K. R. French. 1992. The cross-section of expected stock returns. *The Journal of Finance* 47:427–65.
- Granger, C., E. Maasoumi, and J. Racine. 2004. A dependence metric for possibly nonlinear processes. *Journal of Time Series Analysis* 25:649–69.
- Gul, F. 1991. A theory of disappointment aversion. *Econometrica* 59:667–86.
- Hall, P. 1992. The bootstrap and edgeworth expansion. *Springer series in statistics* .
- Hansen, L. P., and R. Jagannathan. 1991. Implications of security market data for models of dynamic economies. *The Journal of Political Economy* 99:225–62.
- Harvey, C. R., and A. Siddique. 2000. Conditional skewness in asset pricing tests. *The Journal of Finance* 55:1263–95.
- Hong, Y., J. Tu, and G. Zhou. 2007. Asymmetries in stock returns: Statistical tests and economic evaluation. *Review of Financial Studies* 20:1547–81.
- Hong, Y., and H. White. 2005. Asymptotic distribution theory for nonparametric entropy measures of serial dependence. *Econometrica* 73:837–901.

- Horowitz, J. L. 2001. The bootstrap. *Handbook of econometrics* 5:3159–228.
- Jegadeesh, N., and S. Titman. 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance* 48:65–91.
- Kelly, B., and H. Jiang. 2014. Tail risk and asset prices. *Review of Financial Studies* 27:2841–71.
- Kroner, K., and V. K. Ng. 1998. Modeling asymmetric comovements of asset returns. *Review of Financial Studies* 11:817–44.
- Kullback, S., and R. Leibler. 1951. On information and sufficiency. *The Annals of Mathematical Statistics* 22:79–86.
- Lettau, M., M. Maggiori, and M. Weber. 2014. Conditional risk premia in currency markets and other asset classes. *Journal of Financial Economics* 114:197–225.
- Li, Q., and J. S. Racine. 2007. *Nonparametric econometrics: Theory and practice*. Princeton University Press.
- Maasoumi, E., and J. Racine. 2002. Entropy and predictability of stock market returns. *Journal of Econometrics* 107:291–312.
- Maasoumi, E., and J. S. Racine. 2008. A robust entropy-based test of asymmetry for discrete and continuous processes. *Econometric Reviews* 28:246–61.
- Parzen, E. 1962. On estimation of a probability density function and mode. *The Annals of Mathematical Statistics* 33:1065–76.
- Pástor, L., and R. F. Stambaugh. 2003. Liquidity risk and expected stock returns. *Journal of Political Economy* 111:642–85.
- Patton, A., D. N. Politis, and H. White. 2009. Correction to “automatic block-length selection for the dependent bootstrap” by d. politis and h. white. *Econometric Reviews* 28:372–5.

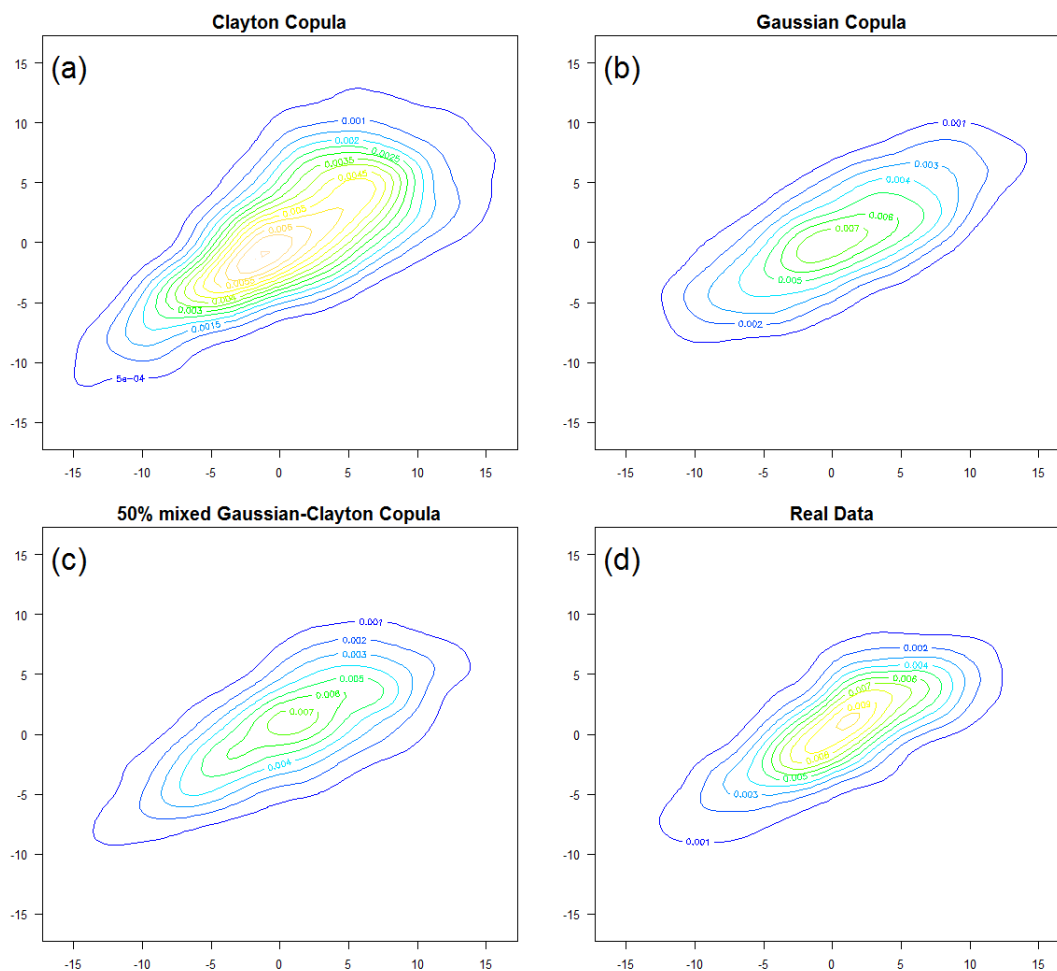
- Racine, J. S., and E. Maasoumi. 2007. A versatile and robust metric entropy test of time-reversibility, and other hypotheses. *Journal of Econometrics* 138:547–67.
- Rosenblatt, M. 1956. Remarks on some nonparametric estimates of a density function. *The Annals of Mathematical Statistics* 27:832–7.
- Schwert, G. W. 1989. Why does stock market volatility change over time? *The Journal of Finance* 44:1115–53.
- Segal, G., I. Shaliastovich, and A. Yaron. 2015. Good and bad uncertainty: Macroeconomic and financial market implications. *Journal of Financial Economics* 117:369–97.
- Shannon, C. 1948. A mathematical theory of communication. *Bell System Technical Journal* 27:379–423.
- Skaug, H. J., and D. Tjøstheim. 1993. A nonparametric test of serial independence based on the empirical distribution function. *Biometrika* 80:591–602.
- Tawn, J. A. 1988. Bivariate extreme value theory: models and estimation. *Biometrika* 75:397–415.
- Tjøstheim, D. 1996. Measures of dependence and tests of independence. *Statistics: A Journal of Theoretical and Applied Statistics* 28:249–84.
- Ullah, A. 1996. Entropy, divergence and distance measures with econometric applications. *Journal of Statistical Planning and Inference* 49:137–62.
- Zhou, G. 1993. Asset-pricing tests under alternative distributions. *The Journal of Finance* 48:1927–42.

Figure 1: Illustrative Example: symmetric correlation with asymmetric comovement



The figure shows contour plots of the joint distributions of two random variables  $X$  and  $Y$ . In the first quadrant, the joint distribution is a bivariate normal distribution with correlation 0.5. In the third quadrant the distribution is a mixture of two bivariate normal distributions where the mixing parameter is set to 0.5. All normal distributions employed here have unit variances. The means of the normal distributions in the first quadrant are set to 3.5, while those in the third quadrant are set to  $-3.5$ . The points  $(X_0, Y_0)$  and  $(-X_0, -Y_0)$  are symmetric about the origin.

Figure 2: Copula dependence structures and contour plots



This figure shows contour plots of random samples simulated using Copula–GARCH models. The value-weighted returns of the smallest size portfolio and the market portfolio are used as the base asset to fit parameters for the data generate process. The sub-figures are labeled as follows: (a) a Clayton copula, (b) a Gaussian copula, and (c) a mixed Gaussian–Clayton copula with mixing weights of 0.5 each, as well as (d) the actual data of the value-weighted returns of the smallest size portfolio and the market. The returns are in percentage points with the horizontal axis denoting the size portfolio returns and the vertical axis denoting the market returns.

Table 1: Size and power: Entropy test and HTZ test

|                                                    | Entropy Test |                    | HTZ Test |                    |
|----------------------------------------------------|--------------|--------------------|----------|--------------------|
|                                                    | C={0}        | C={0, 0.5, 1, 1.5} | C={0}    | C={0, 0.5, 1, 1.5} |
| <b>Panel A: <math>\kappa = 100\%</math> (size)</b> |              |                    |          |                    |
| T = 240                                            | 0.032        | 0.021              | 0.000    | 0.005              |
| T = 420                                            | 0.031        | 0.029              | 0.000    | 0.000              |
| T = 600                                            | 0.044        | 0.037              | 0.000    | 0.000              |
| T = 840                                            | 0.055        | 0.047              | 0.000    | 0.000              |
| <b>Panel B: <math>\kappa = 75\%</math></b>         |              |                    |          |                    |
| T = 240                                            | 0.065        | 0.053              | 0.000    | 0.022              |
| T = 420                                            | 0.120        | 0.078              | 0.001    | 0.008              |
| T = 600                                            | 0.233        | 0.117              | 0.002    | 0.003              |
| T = 840                                            | 0.430        | 0.147              | 0.004    | 0.006              |
| <b>Panel C: <math>\kappa = 50\%</math></b>         |              |                    |          |                    |
| T = 240                                            | 0.278        | 0.215              | 0.059    | 0.099              |
| T = 420                                            | 0.679        | 0.334              | 0.152    | 0.113              |
| T = 600                                            | 0.912        | 0.458              | 0.283    | 0.129              |
| T = 840                                            | 0.991        | 0.611              | 0.482    | 0.188              |
| <b>Panel D: <math>\kappa = 37.5\%</math></b>       |              |                    |          |                    |
| T = 240                                            | 0.513        | 0.329              | 0.169    | 0.185              |
| T = 420                                            | 0.897        | 0.501              | 0.369    | 0.219              |
| T = 600                                            | 0.984        | 0.657              | 0.584    | 0.288              |
| T = 840                                            | 0.999        | 0.809              | 0.778    | 0.434              |
| <b>Panel E: <math>\kappa = 25\%</math></b>         |              |                    |          |                    |
| T = 240                                            | 0.733        | 0.531              | 0.382    | 0.314              |
| T = 420                                            | 0.978        | 0.728              | 0.688    | 0.450              |
| T = 600                                            | 1.000        | 0.856              | 0.855    | 0.580              |
| T = 840                                            | 1.000        | 0.954              | 0.962    | 0.760              |
| <b>Panel F: <math>\kappa = 0\%</math></b>          |              |                    |          |                    |
| T = 240                                            | 0.970        | 0.813              | 0.759    | 0.657              |
| T = 420                                            | 1.000        | 0.971              | 0.953    | 0.838              |
| T = 600                                            | 1.000        | 0.986              | 0.987    | 0.928              |
| T = 840                                            | 1.000        | 0.998              | 0.997    | 0.978              |

The table reports the rejection rates for the null hypothesis of symmetric comovement based on 1,000 Monte Carlo simulations, where the nominal size of the tests is set to 5%. When  $\kappa = 100\%$ , the DGP is a joint normal distribution and the rejection rates are the empirical sizes. In all other cases, the rejection rates reflect empirical power. The Clayton copula parameter  $\tau = 5.768$  and the Normal copula parameter  $\rho = 0.951$ . The specification for the marginal distribution is a standard GARCH(1,1):  $r_{i,t} = \mu_i + \varepsilon_{i,t}$ , where  $\varepsilon_{i,t}$  is normally distributed with a time-varying variance  $\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2$ .

Table 2: Testing Asymmetry

| Panel A: Size           |              |          |                    |              |          |           |         |                    |         |          |
|-------------------------|--------------|----------|--------------------|--------------|----------|-----------|---------|--------------------|---------|----------|
| Portfolios              | Entropy Test |          |                    |              |          | HTZ Test  |         |                    |         | Skewness |
|                         | C={0}        |          | C={0, 0.5, 1, 1.5} |              |          | C={0}     |         | C={0, 0.5, 1, 1.5} |         | Coskew   |
|                         | $S_p$<br>100 | $\times$ | p-Value            | $S_p$<br>100 | $\times$ | Test-stat | p-Value | Test-stat          | p-Value |          |
| Size 1                  | 2.027        |          | 0.010              | 15.483       |          | 4.212     | 0.040   | 9.715              | 0.046   | -0.167   |
| Size 2                  | 1.963        |          | 0.000              | 13.869       |          | 2.049     | 0.152   | 3.281              | 0.512   | -0.262   |
| Size 3                  | 1.868        |          | 0.020              | 12.855       |          | 0.937     | 0.333   | 1.108              | 0.893   | -0.448   |
| Size 4                  | 1.689        |          | 0.013              | 9.506        |          | 0.613     | 0.434   | 2.095              | 0.718   | -0.510   |
| Size 5                  | 1.690        |          | 0.030              | 8.945        |          | 0.431     | 0.512   | 5.015              | 0.286   | -0.524   |
| Size 6                  | 1.596        |          | 0.045              | 10.322       |          | 0.234     | 0.629   | 3.134              | 0.536   | -0.540   |
| Size 7                  | 1.477        |          | 0.065              | 8.343        |          | 0.092     | 0.761   | 0.849              | 0.932   | -0.497   |
| Size 8                  | 1.510        |          | 0.085              | 9.654        |          | 0.099     | 0.753   | 0.146              | 0.997   | -0.470   |
| Size 9                  | 1.695        |          | 0.075              | 10.046       |          | 0.005     | 0.945   | 0.030              | 1.000   | -0.450   |
| Size 10                 | 1.511        |          | 0.055              | 8.638        |          | 0.008     | 0.930   | 0.029              | 1.000   | -0.347   |
| Panel B: Book-to-Market |              |          |                    |              |          |           |         |                    |         |          |
| Portfolios              | Entropy Test |          |                    |              |          | HTZ Test  |         |                    |         | Skewness |
|                         | C={0}        |          | C={0, 0.5, 1, 1.5} |              |          | C={0}     |         | C={0, 0.5, 1, 1.5} |         | Coskew   |
|                         | $S_p$<br>100 | $\times$ | p-Value            | $S_p$<br>100 | $\times$ | Test-stat | p-Value | Test-stat          | p-Value |          |
| B/M 1                   | 1.248        |          | 0.115              | 8.400        |          | 0.023     | 0.880   | 0.341              | 0.987   | -0.230   |
| B/M 2                   | 1.208        |          | 0.085              | 8.649        |          | 0.024     | 0.876   | 0.093              | 0.999   | -0.458   |
| B/M 3                   | 1.003        |          | 0.263              | 8.683        |          | 0.060     | 0.807   | 0.066              | 0.999   | -0.527   |
| B/M 4                   | 1.626        |          | 0.138              | 8.530        |          | 0.064     | 0.800   | 1.829              | 0.767   | -0.474   |
| B/M 5                   | 1.610        |          | 0.055              | 8.068        |          | 0.145     | 0.703   | 2.769              | 0.597   | -0.427   |
| B/M 6                   | 1.815        |          | 0.025              | 8.835        |          | 0.054     | 0.817   | 1.099              | 0.894   | -0.432   |
| B/M 7                   | 1.805        |          | 0.058              | 8.527        |          | 0.082     | 0.774   | 0.590              | 0.964   | -0.137   |
| B/M 8                   | 1.571        |          | 0.098              | 11.316       |          | 0.226     | 0.634   | 2.954              | 0.566   | -0.464   |
| B/M 9                   | 2.162        |          | 0.005              | 12.266       |          | 0.447     | 0.504   | 1.667              | 0.797   | -0.325   |
| B/M 10                  | 1.657        |          | 0.075              | 11.294       |          | 0.805     | 0.370   | 2.133              | 0.711   | 0.069    |
| Panel C: Momentum       |              |          |                    |              |          |           |         |                    |         |          |
| Portfolios              | Entropy Test |          |                    |              |          | HTZ Test  |         |                    |         | Skewness |
|                         | C={0}        |          | C={0, 0.5, 1, 1.5} |              |          | C={0}     |         | C={0, 0.5, 1, 1.5} |         | Coskew   |
|                         | $S_p$<br>100 | $\times$ | p-Value            | $S_p$<br>100 | $\times$ | Test-stat | p-Value | Test-stat          | p-Value |          |
| L                       | 2.758        |          | 0.008              | 6.962        |          | 1.188     | 0.276   | 7.831              | 0.098   | 0.663    |
| 2                       | 1.331        |          | 0.170              | 3.690        |          | 0.309     | 0.578   | 0.980              | 0.913   | 0.224    |
| 3                       | 1.345        |          | 0.080              | 2.095        |          | 0.070     | 0.791   | 0.907              | 0.924   | 0.317    |
| 4                       | 1.974        |          | 0.093              | 8.202        |          | 0.070     | 0.791   | 1.009              | 0.908   | -0.127   |
| 5                       | 1.498        |          | 0.053              | 6.968        |          | 0.069     | 0.793   | 3.081              | 0.544   | -0.278   |
| 6                       | 2.059        |          | 0.013              | 9.794        |          | 0.242     | 0.623   | 1.924              | 0.750   | -0.408   |
| 7                       | 1.614        |          | 0.010              | 10.328       |          | 0.048     | 0.826   | 0.274              | 0.991   | -0.482   |
| 8                       | 1.487        |          | 0.068              | 11.190       |          | 0.167     | 0.683   | 0.388              | 0.983   | -0.360   |
| 9                       | 2.297        |          | 0.000              | 14.095       |          | 0.275     | 0.600   | 3.467              | 0.483   | -0.590   |
| W                       | 2.221        |          | 0.003              | 13.741       |          | 0.749     | 0.387   | 5.384              | 0.250   | -0.431   |

The table reports both the test statistics and the p-values of the entropy test and the HTZ test. The last two columns report skewness and coskewness. The test assets are the value-weighted size, book-to-market and momentum portfolios. The data are monthly and the sample period is from January 1965 to December 2013.

Table 3: Correlations

|           | LQP    | UQP    | DownAsy | $\beta$ | $\beta^-$ | $\beta^+$ | Size   | Bm     | Turn   | Illiq  | Mom    | Ivol   | Coskew | Cokurt | Max |
|-----------|--------|--------|---------|---------|-----------|-----------|--------|--------|--------|--------|--------|--------|--------|--------|-----|
| LQP       | 1      |        |         |         |           |           |        |        |        |        |        |        |        |        |     |
| UQP       | -0.080 | 1      |         |         |           |           |        |        |        |        |        |        |        |        |     |
| DownAsy   | 0.462  | -0.515 | 1       |         |           |           |        |        |        |        |        |        |        |        |     |
| $\beta$   | 0.463  | 0.436  | -0.029  | 1       |           |           |        |        |        |        |        |        |        |        |     |
| $\beta^-$ | 0.300  | 0.273  | -0.036  | 0.800   | 1         |           |        |        |        |        |        |        |        |        |     |
| $\beta^+$ | 0.356  | 0.340  | -0.011  | 0.797   | 0.528     | 1         |        |        |        |        |        |        |        |        |     |
| Size      | 0.299  | 0.473  | -0.079  | 0.222   | 0.097     | 0.233     | 1      |        |        |        |        |        |        |        |     |
| Bm        | -0.071 | -0.207 | 0.066   | -0.259  | -0.212    | -0.202    | -0.280 | 1      |        |        |        |        |        |        |     |
| Turn      | 0.164  | 0.200  | -0.032  | 0.451   | 0.382     | 0.331     | 0.109  | -0.201 | 1      |        |        |        |        |        |     |
| Illiq     | -0.297 | -0.405 | 0.048   | -0.337  | -0.233    | -0.298    | -0.637 | 0.256  | -0.290 | 1      |        |        |        |        |     |
| Mom       | 0.009  | 0.051  | -0.019  | 0.114   | 0.119     | 0.072     | 0.012  | -0.237 | 0.153  | -0.061 | 1      |        |        |        |     |
| Ivol      | -0.091 | -0.203 | 0.029   | 0.308   | 0.336     | 0.170     | -0.525 | -0.058 | 0.232  | 0.357  | 0.042  | 1      |        |        |     |
| Coskew    | 0.033  | 0.024  | 0.038   | 0.013   | -0.343    | 0.388     | 0.038  | -0.010 | -0.025 | 0.010  | -0.040 | 0.007  | 1      |        |     |
| Cokurt    | 0.453  | 0.497  | -0.059  | 0.687   | 0.577     | 0.647     | 0.523  | -0.203 | 0.239  | -0.472 | 0.075  | -0.234 | 0.013  | 1      |     |
| Max       | 0.010  | -0.041 | 0.009   | 0.277   | 0.273     | 0.178     | -0.259 | -0.100 | 0.335  | 0.129  | 0.024  | 0.498  | -0.001 | -0.021 | 1   |

This table reports the the time series averages of the cross-sectional correlations of various risk measures and firm characteristics: the lower quadrant probability (LQP), the upper quadrant probability (UQP), downside asymmetric comovement (DownAsy), the CAPM beta ( $\beta$ ), downside beta ( $\beta^-$ ), upside beta ( $\beta^+$ ), the log of market capitalization (Size), the log of book-to-market ratio (Bm), the turnover ratio (Turn), the normalized Amihud illiquidity measure (Illiq), the past six-month return (Mom), idiosyncratic volatility (Ivol), coskewness (Coskew), cokurtosis (Cokurt), and the maximum daily return over the past one month (Max). The sample covers all U.S. common stocks listed on the NYSE/AMEX/NASDAQ and the sample period is from January 1962 to December 2013.

Table 4: Summary statistics of asymmetry portfolios

| Decile  | DownAsy | $\beta$ | $\beta^-$ | $\beta^+$ | $\beta^- - \beta^+$ | Size  | Bm     | Turn  | Illiq | Mom   | Ivol  | Coskew | Cokurt | Max   |
|---------|---------|---------|-----------|-----------|---------------------|-------|--------|-------|-------|-------|-------|--------|--------|-------|
| 1 Low   | -0.065  | 0.655   | 0.779     | 0.536     | 0.243               | 4.791 | -0.437 | 0.057 | 8.839 | 0.076 | 0.025 | -0.084 | 1.270  | 0.053 |
| 2       | -0.023  | 0.876   | 0.982     | 0.782     | 0.200               | 5.553 | -0.620 | 0.071 | 4.704 | 0.102 | 0.023 | -0.093 | 1.848  | 0.051 |
| 3       | -0.009  | 0.984   | 1.076     | 0.921     | 0.155               | 5.851 | -0.650 | 0.074 | 3.426 | 0.114 | 0.023 | -0.091 | 2.284  | 0.050 |
| 4       | 0.001   | 1.006   | 1.097     | 0.952     | 0.145               | 5.830 | -0.637 | 0.075 | 3.352 | 0.113 | 0.023 | -0.091 | 2.368  | 0.050 |
| 5       | 0.009   | 0.996   | 1.088     | 0.941     | 0.147               | 5.706 | -0.618 | 0.075 | 3.633 | 0.111 | 0.023 | -0.098 | 2.284  | 0.051 |
| 6       | 0.015   | 0.988   | 1.082     | 0.927     | 0.155               | 5.557 | -0.595 | 0.074 | 4.115 | 0.107 | 0.024 | -0.101 | 2.227  | 0.053 |
| 7       | 0.022   | 0.965   | 1.061     | 0.895     | 0.165               | 5.342 | -0.558 | 0.072 | 4.902 | 0.101 | 0.025 | -0.098 | 2.083  | 0.054 |
| 8       | 0.031   | 0.928   | 1.033     | 0.843     | 0.191               | 5.051 | -0.510 | 0.070 | 6.051 | 0.097 | 0.026 | -0.092 | 1.851  | 0.056 |
| 9       | 0.047   | 0.849   | 0.958     | 0.747     | 0.211               | 4.731 | -0.428 | 0.064 | 7.667 | 0.083 | 0.027 | -0.087 | 1.573  | 0.057 |
| 10 High | 0.107   | 0.677   | 0.775     | 0.570     | 0.205               | 4.557 | -0.298 | 0.052 | 9.466 | 0.063 | 0.026 | -0.075 | 1.245  | 0.053 |

The table reports various risk measures and firm characteristics of deciles sorted based on their realized DownAsy each month: downside asymmetric comovement (DownAsy), the CAPM beta ( $\beta$ ), downside beta ( $\beta^-$ ), upside beta ( $\beta^+$ ), the log of market capitalization (Size), the log of book-to-market ratio (Bm), the turnover ratio (Turn), the normalized Amihud illiquidity measure (Illiq), the past six-month return (Mom), idiosyncratic volatility (Ivol), coskewness (Coskew), cokurtosis (Cokurt), and the maximum daily return over the past one month (Max). The sample covers all U.S. common stocks listed on the NYSE/AMEX/NASDAQ and the sample period is from January 1962 to December 2013.

Table 5: Portfolios sorted by asymmetry measures

| Panel A: Downside asymmetry and conventional asymmetry measures |          |               |          |               |           |               |            |               |
|-----------------------------------------------------------------|----------|---------------|----------|---------------|-----------|---------------|------------|---------------|
| DownAsy                                                         |          |               | DownCorr |               | $\beta^-$ |               | Coskewness |               |
| Portfolio                                                       | Return   | Carhart-alpha | Return   | Carhart-alpha | Return    | Carhart-alpha | Return     | Carhart-alpha |
| 1 Low                                                           | 0.652    | 0.022         | 0.693    | 0.090         | 0.772     | 0.161         | 0.956      | 0.101         |
| 2                                                               | 0.746    | 0.060         | 0.790    | 0.116         | 0.837     | 0.157         | 0.948      | 0.171         |
| 3                                                               | 0.849    | 0.122         | 0.892    | 0.152         | 0.886     | 0.141         | 0.863      | 0.138         |
| 4                                                               | 0.943    | 0.160         | 0.908    | 0.116         | 0.909     | 0.115         | 0.774      | 0.076         |
| 5 High                                                          | 1.027    | 0.207         | 0.937    | 0.102         | 0.817     | 0.002         | 0.680      | 0.090         |
| 5 High - 1 Low                                                  | 0.375*** | 0.185**       | 0.244*** | 0.012         | 0.045     | -0.159        | -0.276***  | -0.011        |
| t-stat                                                          | (4.39)   | (2.47)        | (3.35)   | (0.21)        | (0.21)    | (-1.36)       | (-2.70)    | (-0.12)       |

| Panel B: Alternative entropy measures |          |               |               |               |
|---------------------------------------|----------|---------------|---------------|---------------|
| BBC co-entropy                        |          |               | CC co-entropy |               |
| Portfolio                             | Return   | Carhart-alpha | Return        | Carhart-alpha |
| 1 Low                                 | 0.679    | 0.081         | 0.682         | 0.083         |
| 2                                     | 0.776    | 0.084         | 0.778         | 0.085         |
| 3                                     | 0.884    | 0.108         | 0.884         | 0.110         |
| 4                                     | 0.912    | 0.114         | 0.908         | 0.107         |
| 5                                     | 0.949    | 0.155         | 0.948         | 0.156         |
| 5 High - 1 Low                        | 0.269*** | 0.074         | 0.266***      | 0.073         |
| t-stat                                | (4.34)   | (1.41)        | (4.23)        | (1.36)        |

This table reports the equal-weighted average returns and alphas, in percentage points, for stock portfolios sorted by different realized asymmetry measures. DownAsy is the downside asymmetry measure, DownCorr is the downside asymmetric correlation, and  $\beta^-$  is the downside beta. BBC and CC co-entropies are based on Backus, Boyarchenko and Chernov (2016) and Chabi-Yo and Colacito (2016), respectively. The row labeled “5 High - 1 Low” reports the difference between the returns/alphas of portfolios 5 and 1 as well as the associated  $t$ -statistics. The sample covers all U.S. common stocks listed on the NYSE/AMEX/NASDAQ for the period from January 1963 to December 2013. \*, \*\*, and \*\*\* indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.

Table 6: Determinants of Time-Varying Premium of DownAsy

|              | (1)                               | (2)                | (3)                | (4)                | (5)                                  | (6)                | (7)                 | (8)                 |
|--------------|-----------------------------------|--------------------|--------------------|--------------------|--------------------------------------|--------------------|---------------------|---------------------|
|              | Premium based on realized DownAsy |                    |                    |                    | Premium based on MA smoothed DownAsy |                    |                     |                     |
| Mkt Vol      | -0.124***<br>(-2.83)              |                    |                    | -0.087*<br>(-1.78) | -0.118***<br>(-3.05)                 |                    |                     | -0.086**<br>(-1.99) |
| Liq          |                                   | 4.062***<br>(3.01) |                    | 2.963*<br>(1.93)   |                                      | 3.269***<br>(2.72) |                     | 2.316*<br>(1.71)    |
| BWsent       |                                   |                    | -0.036<br>(-0.39)  | -0.043<br>(-0.47)  |                                      |                    | -0.166**<br>(-2.04) | -0.170**<br>(-2.10) |
| Constant     | 0.494***<br>(5.22)                | 0.497***<br>(5.29) | 0.381***<br>(4.26) | 0.561***<br>(5.42) | 0.419***<br>(4.99)                   | 0.404***<br>(4.83) | 0.301***<br>(3.81)  | 0.459***<br>(5.03)  |
| Observations | 612                               | 612                | 582                | 582                | 612                                  | 612                | 582                 | 582                 |
| R-squared    | 0.013                             | 0.015              | 0.000              | 0.020              | 0.015                                | 0.012              | 0.007               | 0.026               |

The table reports various regressions where the dependent variables are the asymmetry premium (realized or moving average smoothed), and the independent variables include market volatility, market liquidity, and investor sentiment. Mkt Vol is the monthly variance of the daily value-weighted market return. Aggregate stock market liquidity (Liq) is provided by Pastor and Stambaugh (2003). The sample period is from January 1963 to December 2013. BWsent denotes the sentiment index of Baker and Wurgler (2006), available only since July 1965.

Table 7: Portfolios with double sorts

| <b>Panel A: DownCorr and DownAsy</b> |                |          |          |          |                 |          |
|--------------------------------------|----------------|----------|----------|----------|-----------------|----------|
| Portfolio                            | 1 Low DownCorr | 2        | 3        | 4        | 5 High DownCorr | Average  |
| 1 Low DownAsy                        | 0.594          | 0.660    | 0.615    | 0.686    | 0.738           | 0.659    |
| 2                                    | 0.527          | 0.630    | 0.804    | 0.877    | 0.893           | 0.746    |
| 3                                    | 0.704          | 0.804    | 0.879    | 0.938    | 0.950           | 0.855    |
| 4                                    | 0.763          | 0.890    | 1.045    | 0.995    | 1.053           | 0.949    |
| 5 High DownAsy                       | 0.876          | 0.963    | 1.116    | 1.045    | 1.049           | 1.010    |
| 5 High - 1 Low                       | 0.282***       | 0.303*** | 0.501*** | 0.359*** | 0.311***        | 0.351*** |
| t-stat                               | (3.04)         | (3.31)   | (4.73)   | (3.41)   | (2.93)          | (4.33)   |

| <b>Panel B: <math>\beta^-</math> and DownAsy</b> |                 |          |          |          |                  |          |
|--------------------------------------------------|-----------------|----------|----------|----------|------------------|----------|
| Portfolio                                        | 1 Low $\beta^-$ | 2        | 3        | 4        | 5 High $\beta^-$ | Average  |
| 1 Low DownAsy                                    | 0.656           | 0.670    | 0.727    | 0.732    | 0.608            | 0.679    |
| 2                                                | 0.674           | 0.761    | 0.763    | 0.808    | 0.698            | 0.741    |
| 3                                                | 0.805           | 0.852    | 0.885    | 0.908    | 0.849            | 0.860    |
| 4                                                | 0.885           | 0.926    | 0.950    | 0.969    | 0.885            | 0.923    |
| 5 High DownAsy                                   | 0.837           | 0.975    | 1.105    | 1.123    | 1.040            | 1.016    |
| 5 High - 1 Low                                   | 0.181**         | 0.305*** | 0.379*** | 0.391*** | 0.432***         | 0.337*** |
| t-stat                                           | (2.23)          | (3.66)   | (4.25)   | (3.70)   | (3.25)           | (4.55)   |

| <b>Panel C: Coskewness and DownAsy</b> |              |          |          |          |               |          |
|----------------------------------------|--------------|----------|----------|----------|---------------|----------|
| Portfolio                              | 1 Low Coskew | 2        | 3        | 4        | 5 High Coskew | Average  |
| 1 Low DownAsy                          | 0.685        | 0.756    | 0.619    | 0.580    | 0.557         | 0.640    |
| 2                                      | 0.928        | 0.829    | 0.771    | 0.655    | 0.509         | 0.739    |
| 3                                      | 0.962        | 0.959    | 0.863    | 0.783    | 0.719         | 0.857    |
| 4                                      | 1.084        | 1.108    | 1.066    | 0.878    | 0.802         | 0.988    |
| 5 High DownAsy                         | 1.116        | 1.085    | 0.993    | 0.971    | 0.808         | 0.995    |
| 5 High - 1 Low                         | 0.431***     | 0.329*** | 0.374*** | 0.391*** | 0.251***      | 0.355*** |
| t-stat                                 | (3.75)       | (3.18)   | (3.61)   | (4.17)   | (2.67)        | (4.35)   |

This table reports the equal-weighted average returns, in percentage points, of portfolios double-sorted by a realized asymmetry measure first and then by DownAsy: DownCorr (Panel A),  $\beta^-$  (Panel B), and coskewness (Panel C), respectively. DownAsy is defined in equation (16). DownCorr is defined in equation (17).  $\beta^-$  and coskewness are defined in equations (A-2) and (A-3). The row labeled “5 High - 1 Low” reports the difference between the returns of portfolios 5 and 1 in each first-sort quintile, as well as the corresponding  $t$ -statistics. The column labeled “Average” reports the average return of stocks in each second-sort quintile. The sample covers all U.S. common stocks listed on the NYSE/AMEX/NASDAQ and the sample period is from January 1963 to December 2013.  $t$ -statistics are reported in parentheses. \*, \*\*, and \*\*\* indicate significance at the 0.1, 0.05, and 0.01 levels, respectively.